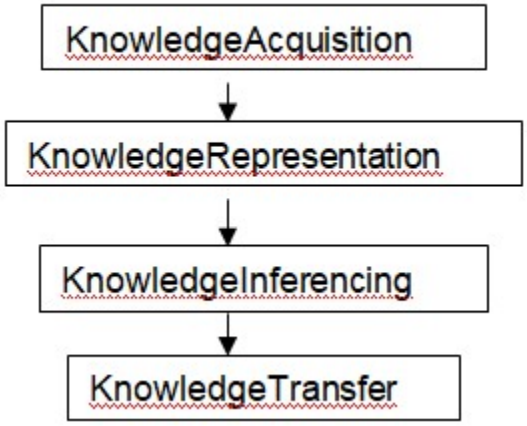
Expert System

- An expert system is a computer system whose performance is guided by specific, expert knowledge in solving problems.
- It is a computer system that simulates the decision-making process of a human expert in a specific domain.
- An expert system is one of the early (large-scale) successes of artificial intelligence.
- An expert system is an "intelligent" program that solves problems in a narrow problem area by using high-quality, specific knowledge rather than an algorithm.
- Expert systems are used by most of the large or medium-sized organizations as a major tool for improving productivity and quality.
- An expert system's knowledge is obtained from expert sources and coded in a form suitable for the system to use in its process.

Steps in Expert System Development



Knowledge Acquisition:

- Knowledge acquisition is the process used to define the rules and ontologies required for a knowledge-based system.
- This includes finding and interviewing domain experts and capturing their knowledge via rules, objects, and frame-based ontologies.
- The expert sources can be domain specialists, articles, journals, databases, etc.

Knowledge Representation:

- Representing information about the world in a form that a computer system can utilize to solve complex tasks.
- It is a set of ontological commitments, i.e., an answer to a question.
- Knowledge representation can be logical representation or structured representation.

- Representing the rules and constraints is done through logic rules, formally known as knowledge representation.

Knowledge Inferencing:

- Knowledge inference refers to acquiring new knowledge from existing facts based on certain rules and constraints.
- Mostly rule-based reasoning (Forward chaining/Backward chaining) is used for inferencing.

Knowledge Transfer:

- Knowledge transfer is the practical problem of transferring knowledge from one part of the organization to another.
- It seeks to organize, create, capture, or distribute knowledge and ensure its availability for future users.

Features of an Expert System

- Should have reasoning capacity: reasoning may be goal-driven (backward chaining) or data-driven (forward chaining).
- Should cope with uncertainty.
- Should have proper knowledge representation, i.e., use knowledge rather than data.
- Should use symbolic representation for knowledge.
- Should use meta-knowledge.
- Should have a user interface.
- Should have the ability to explain solutions with respect to specific problems.

Advantages of Expert Systems

- Provides consistent answers for repetitive decisions, processes, and tasks.
- Holds and maintains a significant level of information.
- Encourages organizations to clarify the logic of their decision-making.
- Asks questions like human expertise.

Disadvantages of Expert Systems

- Lacks common sense needed in some decision-making.
- Cannot make creative responses as a human expert would in unusual circumstances.
- Errors may occur in the knowledge base and lead to wrong decisions.
- Cannot adapt to changing environments unless the knowledge base is changed.

Applications

- Business
- Manufacturing
- Medicine
- Engineering
- Applied science
- Military
- Space
- Transportation
- Education
- Image analysis
- Chemical structure
- Agriculture
- Robotics and many more

History of Expert Systems

- A breakthrough in the field of AI was achieved in the area of Expert Systems.
- **1966** - DENDRAL was developed at Stanford: Analyzes mass spectrographic and NMR data, inferring possible structures of unknown chemicals.
- **1971** - MYCIN was developed at Stanford: Diagnoses and treats infectious blood diseases.
- **1980** - XCON was developed at CMU: One of the first commercial expert systems.
- PROSPECTOR: Analyzing geological data.
- GASOIL: Designing gas-oil separation systems for offshore oil platforms.

Advantages/Disadvantages (Comparison between Human Expert and Expert System)

Human Expert	Expert System
Slow processing	Fast processing
Slow reproduction	Quick replication
Broad focus	Narrow focus
Adaptive needs	Instruction

Features of an Expert System

- Goal-driven reasoning (backward chaining) or data-driven reasoning (forward chaining)
- Coping with uncertainty
- Data representation
- User interface
- Ability to explain solutions with respect to specific problems

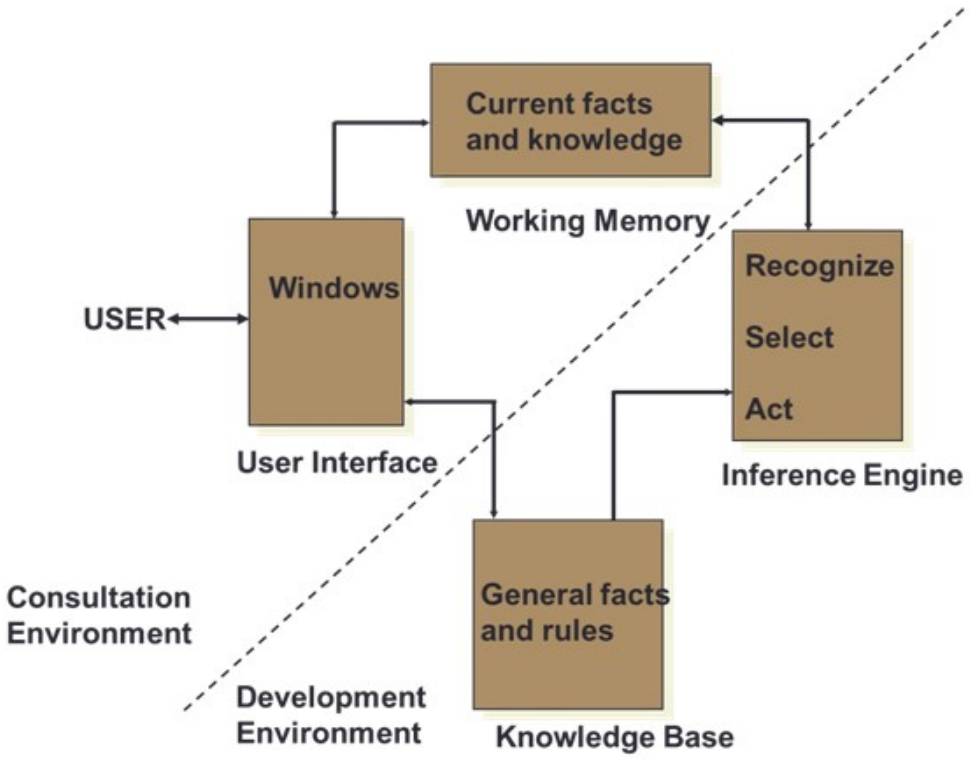
- Use knowledge rather than data
- Use symbolic representation for knowledge
- Should have meta-knowledge

Components of an Expert System

For a workable expert system, more than a set of rules is required. We need four structures:

- **Knowledge Base**: A data structure that contains the rules, often in IF-THEN format.
- **Working Memory**: A data structure that stores information about a specific problem.
- **Inference Engine**: A set of procedures for matching the knowledge base with problem-specific data in working memory.
- **User Interface**: Controls the dialogue between the user and the system.

General Architecture of Expert Systems

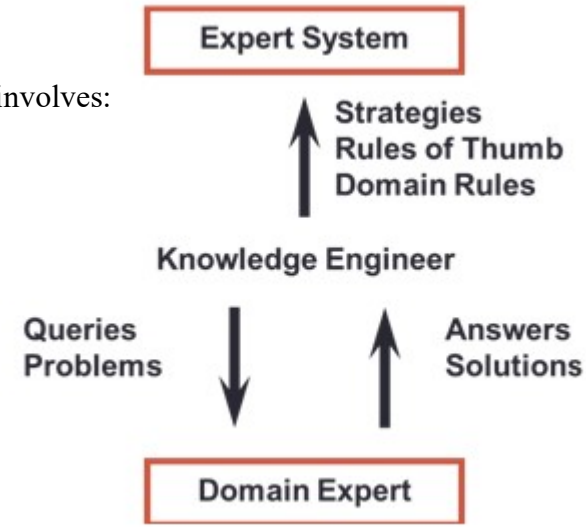


Knowledge in expert systems is usually implemented as a rule:

- A rule is an IF-THEN type statement:
    - IF <certain conditions are true> THEN <take certain actions>.

- As a result, expert systems are sometimes called rule-based systems or production systems.

Knowledge Engineering



The process of filling in an expert system's knowledge base involves:

- Interpreted
- Categorized
- Applied, experienced, and revised.

What it is NOT:

- Data, e.g., numbers.
- Information, i.e., interpreted data.

Types of Knowledge

- **Procedural Knowledge**: "Knowing how" procedures; includes step-by-step sequences and how-to types of instructions. For example, a mobile robot that navigates a building contains procedures such as “navigate to room”, “plan a path”.
- **Declarative Knowledge**: "Knowing what"; descriptive representation of knowledge; tells us facts about truths and associations. For example, “there is a positive association between smoking and cancer”.
- **Episodic Knowledge**: Experiential knowledge.
- **Meta-knowledge**: Knowledge about knowledge.

Sources of Knowledge

- **Expert**: Primary source.
- **End Users**: Usually have a good overview of the problem domain and may provide valuable insight during initial investigations.
- **Secondary/Tertiary Experts**: Can provide specialized knowledge on sub-problems and may give rise to conflicting advice.
- **Literature**: Reports, guidelines, books, manuals, etc., providing background and insight in early stages.

Knowledge Base Size

Frederick Hayes-Roth developed the following heuristics about knowledge bases:

- A convincing demonstration of a knowledge system's power requires about 250 rules.
- An expert level of competence in a narrow area requires about 500 to 1000 rules.
- Expertise in a profession requires about 10,000 rules.

- The limit of human expertise is about 100,000 rules.

Rule Types

- **Relationship (FACT)**: IF the battery is dead THEN the car will not start.
- **Recommendation**: IF the car will not start THEN take a cab.
- **Directive**: IF the car will not start AND the fuel system is okay THEN check out the electrical system.
- **Heuristic**: IF the car will not start AND the car is a 1957 Ford THEN check the float.

Rule Uncertainty

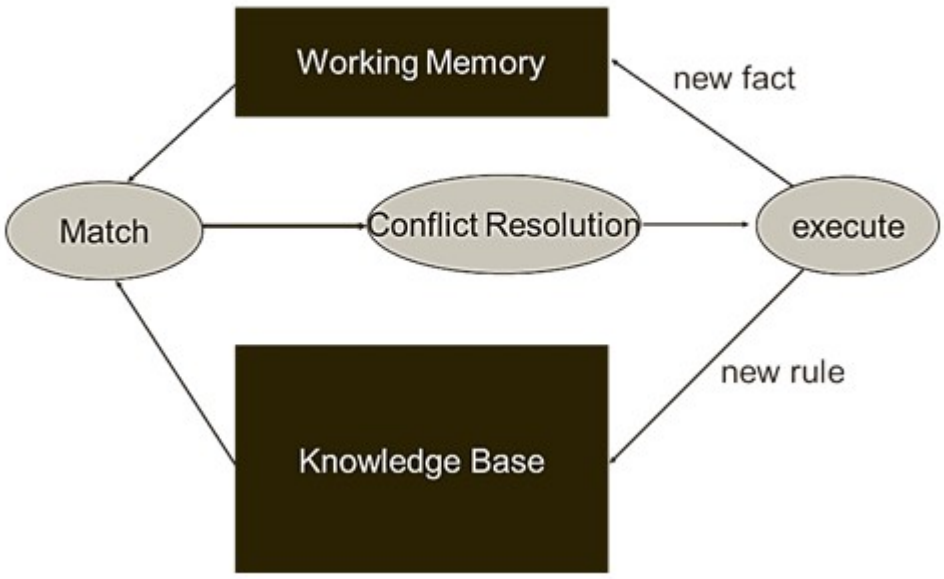
- Vagueness/uncertainty rules: IF inflation is HIGH THEN interest rates might be high.
- Re-write using a confidence quantifier: IF inflation is HIGH THEN interest rates are high (CQ=0.8).

Meta Rules

Rules that express knowledge about how other knowledge should be used:

- IF the car will not start AND the electrical system is operating properly THEN use fuel system rules.

Inference Engine Structure



The inference engine is the mechanism for matching facts with rules and using the results to update the knowledge base. Most inference engines are based on the application of a logical reasoning rule called modus ponens:

- P1: IF A THEN B
- P2: A is true
- Conclude: B is true

Recognize-Select-Act Cycle

It consists of 3 steps:

1. **Match**: Rules are compared to working memory to determine matches.
2. **Conflict Resolution**: Selector enables a single rule for execution.
3. **Execute**: Fire the selected rule.

MYCIN

- An expert system for treating blood infections.
- Diagnoses patients based on reported symptoms and medical test results.
- Could ask for more information and lab test results for diagnosis.
- Recommends a course of treatment; if requested, MYCIN would explain the reasoning that led to its diagnosis and recommendation.
- Uses about 500 production rules; MYCIN operated roughly at the same level of competence as human specialists in blood infections.
- Uses backward chaining for reasoning.

DENDRAL

- The first expert system developed in the late 1960s.
- Designed to analyze mass spectra.
- Based on the mass of fragments seen in the spectra, it would be possible to make inferences about the nature of the molecule tested, identifying functional groups or even the entire molecule.
- Used heuristic knowledge obtained from experienced chemists.
- Uses forward chaining for reasoning.